

ANANYA SRIDHAR

Speech AI & Audio ML Engineer — ASR, TTS & Acoustic Modeling

Mumbai, India · +91 8104921962 · ananyasridhar108@gmail.com

github.com/darayavahoush · vaakify.manaslearning.com · darayavahoush.github.io · linkedin.com/in/ananya-sridhar-ai

PROFESSIONAL SUMMARY

Speech AI and Audio Machine Learning Engineer with hands-on expertise building production ASR, TTS, and phoneme-level acoustic verification pipelines across 8+ Indian languages. Deep experience optimizing Faster-Whisper, Vosk, Coqui TTS, and Kokoro ONNX alongside digital audio signal processing (Librosa, Praat-Parselmouth). Proven track record deploying low-latency clinical speech therapy systems and fintech conversational audio pipelines.

TECHNICAL SKILLS

- **Speech AI & ASR/TTS:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Sarvam AI, Wav2Vec2, Epitran (G2P), Pronunciation Scoring, Forced Alignment
- **Audio DSP & Feature Extraction:** Librosa, Praat-Parselmouth, SoundFile, Spectral Noise Reduction (`noisereduce`), Formant & Jitter/Shimmer Extraction, Pitch Contours
- **AI & Agentic Systems:** LangGraph, LangChain, PyTorch, Hugging Face Transformers, ChromaDB, RAG, Reinforcement Learning
- **Backend & Full-Stack:** Python, C++, FastAPI, WebSockets, PostgreSQL, Docker, Azure Cloud, React.js, Vite

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Speech AI & Systems)

Mar 2026 – Present | Remote / Mumbai, India

- **Multilingual Speech Diagnostic Pipeline:** Architected a pediatric speech assessment engine chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme verification across 8 Indian languages.
- **Phoneme-Level Scoring Engine:** Implemented acoustic feature analysis (formant shifts, duration ratios, and G2P phonetic alignments via Librosa and Epitran) to classify articulatory errors (place, manner, voicing).
- **Adaptive Practice Engine:** Developed a Q-learning difficulty controller dynamically tuning practice games (BreathQuest, VaakMirror, Chime) based on real-time acoustic accuracy metrics.
- **Production Delivery:** Shipped FastAPI REST and WebSocket endpoints supporting real-time streaming audio ingestion with sub-300ms feedback latency.

Nivesh AI — AI/ML Engineer Intern

Jun 2026 – Present | Mumbai, India

- **Fintech Speech Ingestion Pipeline:** Built production audio transcription and summarization pipelines on Azure supporting 10,000+ mutual fund distributors (MFDs).
- **Multilingual STT Optimization:** Evaluated and tuned Sarvam AI and Whisper models with spectral denoising (`noisereduce`), improving transcription fidelity on mixed Indian English/Hindi advisor conversations.
- **Automated Advisory Documentation:** Implemented a dense RAG pipeline parsing conversation transcripts into structured CRM action notes, cutting manual documentation turnaround by 80%.

FEATURED PROJECTS

Vaakify & Chime (PhonemeQuest) — Phoneme Verification & Articulation Engine

[GitHub: Speech-Repeater](#) · [GitHub: PhonemeQuest](#)

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, Docker

- Engineered a phoneme-level articulation engine extracting jitter, shimmer, formant frequencies (F1/F2), and pitch contours from speech, powering gamified practice modules in English, Hindi, and Kannada.

BreathQuest — Pediatric Respiratory & Acoustic Analysis Platform

[Live Demo: breathquest-three.vercel.app](#) · [GitHub](#)

React, Vite, Web Audio API, HTML5 Canvas, FastAPI, PostgreSQL

- Built a gamified respiratory and speech therapy platform with 6 HTML5 canvas mini-games analyzing live microphone amplitude, sustained duration, and frequency modulation via Web Audio API.

VaakBot — Clinical Speech Pathology & Autism Guidance Assistant

[GitHub: VaakBot](#)

Python, FastAPI, Azure OpenAI, Vector Search, RAG, Pydantic

- Built a RAG chatbot on Azure indexing reference clinical manuals and condition taxonomies to guide therapists on pediatric speech intervention hierarchies with zero hallucination.

VaakSiddhi POC — Automated Speech Verification Platform

[GitHub: VaakSiddhi-POC](#)

Python, Librosa, PyTorch, FastAPI, React

- Developed an AI speech verification platform automating acoustic phoneme scoring and progress tracking on tablet form-factors across multilingual Indian settings.

EDUCATION & CERTIFICATIONS

- **Bachelor of Technology (B.Tech),** SASTRA University (Expected 2027)
- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)